

15-2d A Monopoly's Profit

How much profit does a monopoly make? To see a monopoly firm's profit in a graph, recall that profit equals total revenue (TR) minus total costs (TC):

$$\text{Profit} = TR - TC.$$

We can rewrite this as

$$\text{Profit} = (TR/Q - TC/Q) \times Q.$$

TR/Q is average revenue, which equals the price, P , and TC/Q is average total cost, ATC . Therefore,

$$\text{Profit} = (P - ATC) \times Q.$$

This equation for profit (which also holds for competitive firms) allows us to measure the monopolist's profit in our graph.

Consider the shaded box in [Figure 5](#). The height of the box (the segment BC) is price minus average total cost, $P - ATC$, which is the profit on the typical unit sold. The width of the box (the segment DC) is the quantity sold, Q_{MAX} . Therefore, the area of this box is the monopoly firm's total profit.

Figure 5

The Monopolist's Profit

The area of the box BCDE equals the profit of the monopoly firm. The height of the box (BC) is price minus average total cost, which equals profit per unit sold. The width of the box (DC) is the number of units sold.

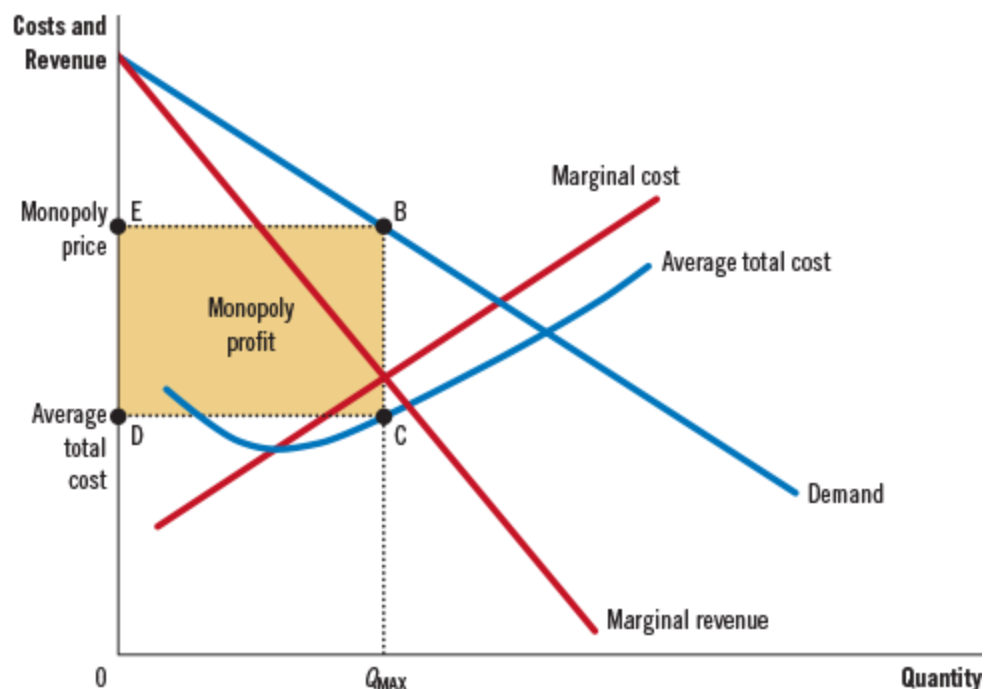


Table 2 summarizes the lessons we have learned so far about how a monopoly firm maximizes profit.

Table 2

Profit-Maximizing Rules for a Monopoly Firm

1. Derive the MR curve from the demand curve.
2. Find Q at which $MR = MC$.
3. On the demand curve, find P at which consumers will buy Q .
4. If $P > ATC$, the monopoly earns a profit.

Case Study: Monopoly Drugs versus Generic Drugs

According to our analysis, prices are determined differently in monopolized markets and competitive markets. A natural place to test this theory is the market for pharmaceutical drugs because this market takes on both market structures. When a firm discovers a new drug, patent laws give the firm a monopoly on the sale of that drug. But eventually, the firm's patent expires, and any company can make and sell the drug. At that time, the market switches from being monopolistic to being competitive.

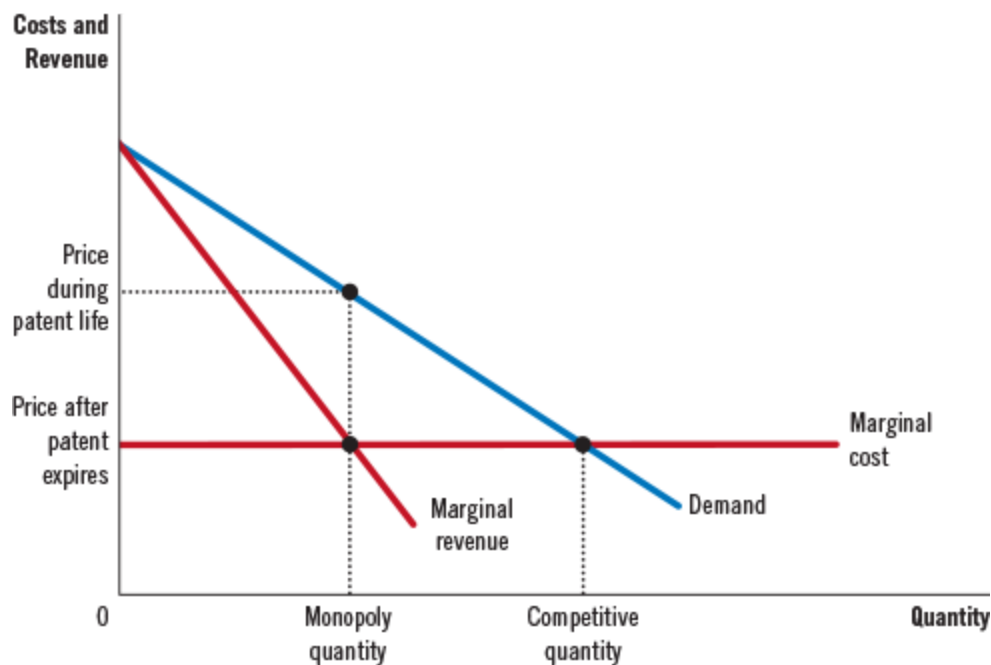
What should happen to the price of a drug when the patent expires? Consider Figure 6, which shows the market for a typical drug. In this figure, the marginal cost of producing the drug is assumed to be constant. (This is roughly true for many drugs.) During the life of the patent, the monopoly firm maximizes profit by producing the quantity at which marginal

revenue equals marginal cost and charging a price well above marginal cost. But when the patent expires, the profit from making the drug should encourage new firms to enter the market. As the market becomes more competitive, the price should fall to equal marginal cost.

Figure 6

The Market for Drugs

When a patent gives a firm a monopoly over the sale of a drug, the firm charges the monopoly price, which is well above the marginal cost of making the drug. When the patent on a drug expires, new firms enter the market, making it more competitive. As a result, the price falls from the monopoly price to marginal cost.



Experience does, in fact, support our theory. When the patent on a drug expires, other companies quickly enter and begin selling generic products that are chemically identical to the former monopolist's brand-name product. Just as our analysis predicts, the competitively produced generic drugs are priced well below the price that the monopolist was charging.

The expiration of a patent, however, does not cause the monopolist to lose all of its market power. Some consumers remain loyal to the brand-name drug, perhaps out of fear that the new generic drugs are not actually the same as the drug they have been using for years. As a result, the former monopolist can continue to charge a price above the price charged by its new competitors.

For example, the drug fluoxetine, an antidepressant taken by millions of Americans, was originally sold under the brand name Prozac. Since the expiration of the patent in 2001, consumers have had a choice of the original drug, Prozac, and generic versions of the same medicine. Yet Prozac today sells for about three times the price of generic fluoxetine. This

price differential persists because some consumers are not convinced that the two pills are perfect substitutes.

FYI

Why a Monopoly Does Not Have a Supply Curve

You may have noticed that we have analyzed the price in a monopoly market using the market demand curve and the firm's cost curves. We have made no mention of the market supply curve. By contrast, when we analyzed prices in competitive markets beginning in [Chapter 4](#), the two most important words were always *supply* and *demand*.

What happened to the supply curve? Although monopoly firms make decisions about what quantity to supply, a monopoly does not have a supply curve. A supply curve tells us the quantity that firms choose to supply at any given price. This concept makes sense when we are analyzing competitive firms, which are price takers. But a monopoly firm is a price maker, not a price taker. It is not meaningful to ask what amount such a firm would produce at any given price because it does not take the price as given. Instead, when the firm chooses the quantity to supply, that decision—along with the demand curve—determines the price.

Indeed, the monopolist's decision about how much to supply is impossible to separate from the demand curve it faces. The shape of the demand curve determines the shape of the marginal-revenue curve, which in turn determines the monopolist's profit-maximizing quantity. In a competitive market, each firm's supply decisions can be analyzed without knowing the demand curve, but the same is not true in a monopoly market. Therefore, we never talk about a monopoly's supply curve.

QuickQuiz

3. For a profit-maximizing monopoly that charges a single price, what is the relationship between price P , marginal revenue MR , and marginal cost MC ?
- a. $P = MR$ and $MR = MC$.
 - b. $P > MR$ and $MR = MC$.
 - c. $P = MR$ and $MR > MC$
 - d. $P > MR$ and $MR > MC$.

4. If a monopoly's fixed costs increase, its price will _____ and its profit will _____ .

- a. increase; decrease
- b. decrease; increase
- c. increase; stay the same
- d. stay the same; decrease

Chapter 15: Monopoly: 15-2d A Monopoly's Profit

Book Title: WGU Economics

Printed By: Mike Benton (mbento6@wgu.edu)

© 2021 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.